

RESEARCH ARTICLE

Hi-Le and HiTCLe: Ensemble Learning Approaches for Early Diabetes Detection Using Deep Learning and Explainable Artificial Intelligence

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ABSTRACT Diabetes is a metabolic disease caused by the body's failure to use insulin or break down meals correctly. Every year, an alarming number of new cases of diabetes are recorded. A poor lifestyle and an unfavorable environment are the two main causes of diabetes. If it is not treated at early stages, it becomes a lifelong disease and further leads to failure of important organs such as the kidneys, heart, eyes, and so on. This danger can be decreased with timely and precise identification. Deep Learning (DL) is the best method for illness prediction, as demonstrated by recent developments in DL for clinical use. We have proposed two ensemble learning approaches: blending and hybrid by using the Diabetes Prediction Dataset (DPD), which is a highly imbalanced dataset. The number of diabetic patients in it are 8500 whereas, the number of non-diabetic individuals are 91500. To overcome the class imbalance problem, a Proximity-Weighted Synthetic Oversampling (ProWSyn) technique is implemented. We have proposed a hybrid of highway and LeNet model, named Hi-Le, for early and accurate diabetes detection. Hi-Le model achieves an accuracy of 94%, a F1-Score of 96%, precision score of 94% and recall of 95% and beats its individual models in terms of accuracy, F1-Score, precision and recall. We have also proposed a blending model named HiTCLe using Highway, LeNet, and a Temporal Convolutional Network (TCN) to detect and predict diabetes at an early stage. HiTCLe performs best, beats its individual models, highway, TCN and LeNet, and achieves an accuracy score of 94% and a F1-Score of 94%, whereas individual models achieve an accuracy score between 89% and 91% on 10 epochs. To validate models' results, we have implemented K-Fold Cross Validation (K-FCV). Also, to know the features contributions, we have implemented Shapley Additive exPlanations (SHAP) post processing technique. Both ensemble learning models outperform their individual models in term of accurate diabetes detection and prediction.

INDEX TERMS Deep learning, diabetes prediction, hybrid model, LeNet, proximity-weighted synthetic oversampling, SHapley additive exPlanations, 10-fold cross-validation.

I. INTRODUCTION

Diabetes, a metabolic disease, was common in adults in old time. However, now a days, it is increasing in young people also. It is due to their busy life style as they are not able to maintain their health properly. The food we eat

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is the main source of absorption of glucose. Glucose is created and then transported throughout the body via the blood. A part of this glucose is used by the brain to support healthy cognitive function and clear thinking. The body's cells and the liver, which serves as the body's glucose reserve, receive the remainder of glucose. In order to keep blood sugar levels in the bloodstream stable and continuous, the liver is essential. It stores energy in a form that the body may use when needed, acting as a storehouse and producer of glucose. Insulin and glucagon are the two hormones that control the storage and release of glucose. The pancreas secretes insulin, which is essential for using stored energy. On the other hand, diabetes arises when the pancreas is unable to make enough insulin, which causes an excess of sugar or glucose to build up in the blood. Diabetes mellitus, another name for the condition, is the accumulation of blood glucose or sugar in the blood or urine to an abnormally high level. It is a chronic disease and its effects are long lasting on various body parts. It leads to kidney failure, arthritis, lower limb amputation, blindness and heart attack. Symptoms of diabetes are as follows [1]:

- Increased hunger
- Frequent infections and slow healing
- Unexplained weight gain or loss
- Joint pain
- Blurry vision
- Poor Dental Health
- Frequent urination

Statistics show that 452 million individuals suffered worldwide from diabetes in 2017, and by 2045, it is predicted that this will increase to 694 million [2]. There are three types of diabetes. In type I diabetes, the pancreas is detected as harmful by the immune system and cannot produce enough insulin according to body requirements. This problem usually occurs in adults and appears after a long time. In type II diabetes, insulin cannot be properly used for energy generation. Type III diabetes, also known as gestational diabetes, is common in pregnant women [3]. There are no proper medications for diabetes cure until it is detected in its early stages. If diabetes is detected in its early stages, it can be managed easily through a healthy lifestyle as, illustrated in Figure 1. In early times, disease was diagnosed by manually reporting, which was an error-prone and an unreliable method. Doctors needed an accurate and early detection system for diabetes detection to save the lives of individuals from destruction.

With the advancement in Artificial Intelligence (AI) and Data Mining (DM), researchers can provide aid to doctors and medical professionals in the early detection of diabetes. A lot of research has been conducted on this topic using Machine Learning (ML) and DL approaches on big data collected from hospitals or from individuals through surveys available on the Internet. ML techniques used in prior studies for diabetes detection included Multi-Layer Perceptron (MLP) [4], Random Forest (RF) [6], [8], [11], [16], Extra Trees (ET) [6], [10], K-Nearest Neighbour (KNN) [2], [7], [11],



FIGURE 1. Diabetes management.

[14], [16], Support Vector Machine (SVM) [8], [9], [11], [16], Deep Neural Network (DNN) [7], Decision Tree (DT) [8], [10], [15], Logistic Regression (LR) [9], [14], [16], Naive Bayes (NB) [2], [8], Adaptive Boosting (AdaBoost) [8], [9], [10], and Gradient Boosting (GradBoost) [17] and their ensemble models [18]. In the last few years, DL has shown remarkable progress in all fields of life, especially healthcare diagnosis. Convolutional Neural Network (CNN) [5], Recurrent Neural Network (RNN) [5], [12], Long Short Term Memory (LSTM) [5], [12], and Gated Recurrent Unit (GRU) [5] and their hybrid learning approaches [13] have been used in previous research for accurate and early diabetes prediction.

A. MAJOR CONTRIBUTIONS

Our main contributions in this study are as follows.

- We have proposed a **hybrid DL model, Hi-Le**, by combining two DL models, Highway and LeNet, for accurate and early detection of diabetes.
- We have proposed a **blending model, HiTCLC**, by combining three DL models, Highway, TCN, and LeNet, for early and reliable detection of diabetes.
- We have used **DPD** which is a highly imbalanced dataset used for training Hi-Le and HiTCLC models.
- To make a balance between the class distribution of the dataset, **ProWSyn** has been used.
- To validate the results of Hi-Le and HiTCLC model, we have implemented 10-FCV.
- To learn feature contribution in diabetes detection in a model, **SHAP** has been implemented on HiTCLC and Hi-Le models.
- Also we have conducted **ablation study** on our HiTCLC and Hi-Le models.

The rest of the paper is divided into various sections. The related work is presented in Section II, the proposed approach

is fully described in Section III, the experiment results are included in Section IV, and our works' conclusion is given in Sections V.

II. RELATED WORK

Many studies have been conducted recently on the use of DL and ensemble learning tools and techniques for diabetes prediction. There has been discussion of various datasets, algorithms, and techniques that the researchers have employed to complete their study. Better results have been obtained with the developed models, which can help healthcare providers make data-driven decisions. The most pertinent papers on using ensemble learning techniques to predict diabetes are reviewed in this section.

In [19], the authors used IoT and DL techniques to develop an automated model of IoT and DL-enabled Healthcare Disease Diagnoses (IoTDL-HDD) using Electrocardiogram (ECG) signals. IoTDL-HDD is able to detect the existence of CVDs in ECG signals using the DL model. To extract useful and important features from ECG signals, the IoTDL-HDD model used Bi-LSTM as a feature extraction technique. A hyperparameter optimizer, the Artificial Flora Optimizer (AFO), is used to improve the efficiency of the Bi-LSTM model. For assigning accurate class labels to ECG signals, a Fuzzy Deep Neural Network (FDNN) model is employed. The proposed model achieves high accuracy of 93% on ECG signals data. In [20], the authors offer a physics-guided DL network for the functional assessment of CVDs using the IoT to address this problem. To extract important and useful features by taking into account the importance of coronary artery features and artery segments, an attentive network was developed. Physical knowledge related to blood flow was used to get explainable functional assessment to loss function. It was guaranteed that functional evaluation must be followed with physical rules. On a synthetic dataset as well as a clinical dataset from the real world, extensive tests were conducted. The findings demonstrated the ability of this method to produce assessments that were valid and physically consistent. Additionally, this approach encourages a wider adoption of DL and IoT in the context of smart health. In [21], to diagnose Gastrointestinal Infections (GI), authors provided an improvement by reducing GI diagnosis time and assessing severity levels to provide benefits to patients as well as clinical staff and doctors. The study uses KVASIR dataset of GI track images was used to train different variants including fine-tuned and simple models of CNN like VGG16, InceptionV3 and Residual Network (ResNet-50). The database includes images of three different GI disorders, including esophagitis, ulcerative colitis, polyps, and also contains healthy colon images. To improve model performance, techniques for enhancing data with statistical metrics were utilized, and the model was evaluated on 1200 images. The proposed ResNet50 model surpasses all other existing systems by attaining accuracy score of 99%.

In [22] authors highlights one of the most common and dangerous illnesses affecting people on a global scale, Chronic Kidney Disease (CKD), fails to appear itself until the kidneys of an individual have been subjected to permanent damage. The growth of CKD is connected with a number of serious side effects, including an increased risk of different illnesses, anemia, hyperlipidemia, damage to the nervous system, challenges during pregnancy, and even severe kidney failure. CKD kills the lives of millions of individuals each year. Since there are no significant symptoms that can be used as a benchmark to identify CKD, diagnosing the condition might be difficult. Sometimes, data may be interpreted wrongly when diagnosis is persistent. The authors proposed a MLP classifier built on DNN to identify CKD in patients. The model was trained on data of 400 people, which took a variety of symptoms and indicators into account. The experiments showed that the model performed with 100% testing accuracy in classification. This paper made use of DL techniques for precisely detecting CKD by learning from dataset attribute reports. The authors proposed a DNN that performed better than traditional ML models like SVM and NB classifiers for predicting CKD achieves 100% accuracy. A stronger alternative to adaptive strategies for categorizing CKD could be neural models. Since they are capable of handling fetched non-linear large data from the dataset. In [23], the authors main goal was to use several ML techniques to forecast the presence of diabetes. Moreover, the prediction accuracy of the aforementioned models was assessed together with other performance measures like precision, recall, and F1 score to identify which model is the most useful in this aspect. RF performed substantially better than the other models, with an accuracy of 82.26%.

In [24], the authors proposed a DL model with an efficient feature selection process for Diabetic Mellitus (DM) prediction. The authors started by gathering IoT based real-time data of patients from Pima Indian Diabetes. The synthetic minority oversampling technique was employed to address the issue of class imbalance and after that balanced data was preprocessed through filling missing data and data standardization. Then, by performing clustering and optimization, the hybrid approach known as k-means clustering-based sailfish optimization was utilized to find the important characteristics from the converted dataset. In the end, the chosen features were supplied into the bi-LSTM with kaiming and swtan incorporated for DM prediction. The proposed approach outperformed current state-of-the-art methodologies, demonstrating its efficacy in predicting DM in real-time IoT information. In [25], the authors wanted to make early prediction of readmissions to keep track of patients who are at high risk of readmission. ML predictive approaches such as ensemble learning using XGBoost, SVM, and ANN has been attempted in an effort to accurately predict whether readmission can occur within 30 days, may never occur, or may occur after 30 days. The author suggested

TABLE 1. Summary of related work for diabetes prediction.

Limitations	Methodology	Evaluation Metrics
IoTDL-HDD model limited to ECG signals data	Developed IoTDL-HDD model using IoT and DL techniques to detect CVDs in ECG signals. Used Bi-LSM for feature extraction, AFO for hyperparameter optimization, and FDNN for accurate class labeling [20]	Accuracy
Limited accuracy for predicting heart disease	Developed physics-guided DL network for functional assessment of CVDs using IoT. Used attentive network for feature extraction and physical knowledge integration into loss function for explainable assessment [21]	Accuracy
Long diagnosis time for GI	Used KVASIR dataset to train CNN models (VGG16, InceptionV3, ResNet-50) for GI diagnosis. Employed statistical metrics for data enhancement. Proposed ResNet50 model achieved 99% accuracy [22]	Accuracy
Difficulty in diagnosing CKD	Proposed MLP classifier built on DNN for CKD identification achieving 100% testing accuracy. Showed superiority of DNN over traditional ML models like SVM and NB classifiers [23]	Accuracy
Low prediction accuracy for diabetes	Used RF for diabetes prediction achieving 82.26% accuracy. Evaluated precision, recall, and F1 score to identify model usefulness [24]	Accuracy
Class imbalance in DM prediction	Proposed DL model with efficient feature selection using IoT data. Employed k-means clustering-based sailfish optimization for feature selection. Achieved efficacy in predicting DM in real-time [25]	Accuracy
High risk of patient readmission	Developed ensemble-based approach (super learner model) for diabetes prediction. Combined LR, DT, RF, GradBoost models. Outperformed base models on three datasets [26]	Accuracy
Interpreting DNN	Proposed hybrid PseKNC, PCA, and DNN model for sequence formulation. Implemented 10-FCV but faced challenges in model interpretability [27]	Accuracy, F1-Score
Feature selection interpretability	Used ensemble learning and Boruta feature selection for diabetes prediction. Faced challenges with unsupervised clustering and feature selection interpretability [28]	Accuracy, F1-Score
Class imbalance in diabetes prediction	Proposed hybrid SMOTE and ENN technique for class imbalance. Validated results through 10-FCV but challenges with dataset features' effectiveness [29]	Accuracy, F1-Score, Specificity
Interpreting intelligent algorithm for diabetes prediction	Developed attention-based GRU algorithm for diabetes prediction. Challenges with algorithm interpretability and sensor reliability [30]	F1-score, Recall, Error Rate, Accuracy, Precision
Reducing training time for diabetes prediction	Employed ANN for diabetes prediction with automatic feature extraction. Achieved classification accuracy but faced challenges with training time [31]	Classification Accuracy

a method to guide the ANN during its gradient descent optimization process by distinguishing between consistent and inconsistent data within each batch. The findings demonstrated that using this model, they have achieved best accuracy in both 2-class and 3-class variations of the tested baseline dataset up to 1.5%. In comparison to normal ANN, driven ANN is also capable of achieving greater error convergence.

The authors in [26] proposed a hybrid Pseudo-K-Tuple Nucleotide Composition (PseKNC) for sequence formulation, an unsupervised Principal Component Analysis (PCA) algorithm for discriminant feature selection, and a Deep Neural Network (DNN) as a classifier. They implemented this model on different datasets and also implemented 10-FCV. However, Interpreting DNNs may be difficult, particularly

when paired with intricate feature extraction techniques like PCA and PseKNC. It is important to comprehend how interpretable the model is, particularly in situations where clear decision-making is required. In [27], the authors used ensemble learning and Boruta feature selection in a diabetes prediction model. The model included the K-Means++ algorithm for unsupervised data clustering, Boruta feature selection, salient feature extraction from datasets, and stacking of an ensemble learning technique for classification. However, for unsupervised data clustering, the K-Means++ technique may not always be the best option, particularly if the underlying data distribution is not well-suited for K-Means. Moreover, the interpretability of the characteristics that are chosen must be taken into account while using Boruta for feature selection. In [28], the

TABLE 2. Mapping table of existing studies with our proposed work.

Paper	Proposed Solution	Limitation in their Study	Our Contribution
[24]	Used RF for diabetes prediction and performance evaluation.	Lack of discussion on feature importance analysis and model explainability.	We propose novel DL architectures, Hi-Le and HiTCLe, and SHAP analysis to enhance feature importance understanding and model explainability for diabetes diagnosis.
[25]	Developed a DL model with feature selection for DM prediction using IoT data.	Limited exploration of class imbalance handling techniques and model explainability.	We employ the ProWSyn technique and SHAP analysis to improve class imbalance handling and model explainability in diabetes prediction.
[26]	Proposed ensemble learning for early readmission prediction using XGBoost.	Lack of discussion on model ensemble diversity and performance comparison.	We propose novel DL architectures, Hi-Le and HiTCLe, employ the ProWSyn technique, and conduct 10-FCV to enhance ensemble models diversity and performance evaluation in diabetes prognosis.
[27]	Developed a hybrid PseKNC and PCA-DNN model for sequence formulation.	Limited exploration of model interpretability and feature selection robustness.	We introduce novel DL architectures, Hi-Le and HiTCLe, employ the ProWSyn technique, and conduct 10-FCV and SHAP analysis to improve model interpretability and feature selection robustness in diabetes diagnosis.
[28]	Used ensemble learning and Boruta for diabetes prediction.	Lack of discussion on feature selection stability and model ensemble diversity.	We propose novel DL architectures, Hi-Le and HiTCLe, employ the ProWSyn technique, and conduct 10-FCV and SHAP analysis to enhance feature selection stability and model ensemble diversity in diabetes detection.
[29]	Proposed a hybrid SMOTE and ENN technique for class imbalance in diabetes prediction.	Limited exploration of data balancing efficiency and model generalizability.	We employ the ProWSyn technique, and conduct 10-FCV and SHAP analysis to improve data balancing efficiency and model generalizability in diabetes prognosis.
[30]	Developed an intelligent algorithm for diabetes management using attention-based GRU units.	Lack of discussion on the scalability of the algorithm for tabular data.	We propose novel DL architectures, Hi-Le and HiTCLe, employ the ProWSyn technique, and conduct 10-FCV and SHAP analysis to optimize model performance and address concerns regarding scalability for tabular data.
[31]	Proposed a super learner model for diabetes prediction using an ensemble of base models.	Limited exploration of feature selection methods and hyper-parameter tuning.	We introduce novel DL architectures Hi-Le and HiTCLe that learns hidden and intricate patterns from data.
[32]	Developed an R-LSTM hybrid model for glucose prediction using CNN and LSTM features.	Lack of model explainability.	We employ the ProWSyn technique, and implement SHAP analysis to enhance glucose prediction accuracy and interpretability of the model.
[33]	Proposed an ANN-based method for diabetes detection using health indicators.	ANN is not able to learn patterns from data.	We introduce novel DL architectures, Hi-Le and HiTCLe, to learn hidden patterns from data also we employ the ProWSyn technique, and conduct 10-FCV and SHAP analysis to improve classification accuracy and reduce training time for diabetes detection models.
[34]	ML interface for diabetes diagnosis using DT, KNN, SVC, and XGB models. Combined SHAP's explanations with XGB model for prediction.	Limited to conventional ML models, lacking DL techniques.	Proposed DL architectures Hi-Le and HiTCLe, surpassing conventional ML methods.
[35]	MLP algorithm based on DNN for early diabetes prediction. Automated DL framework for hyperparameter tuning.	Limited exploration of feature selection and model explainability.	Employed ProWSyn technique, 10-FCV, and SHAP analysis to enhance feature selection and model explainability.
[36]	Used PSO for feature selection and compared ML techniques. DT, RF, and Naïve Bayes provided best accuracy.	Focused only on conventional ML, neglected DL effectiveness.	Proposed DL architectures Hi-Le and HiTCLe, demonstrating superior performance over conventional ML.

authors' main focus was to deal with the class imbalance problem. The authors proposed a hybrid of SMOTE and ENN data balancing techniques and implemented SVM and RF classifiers on data to check model performance on this balancing technique. The results were also validated through 10-FCV method. Although class imbalance is intended to be addressed by the hybrid strategy that combines SMOTE

and ENN, its effect on the overall performance of the model must be carefully assessed. The features of the dataset may have an impact on the effectiveness of the proposed model.

Enabling early diagnosis of abnormal instances of diabetes that require medical treatment was the primary objective in paper [29]. In order to do this, sensors that can establish

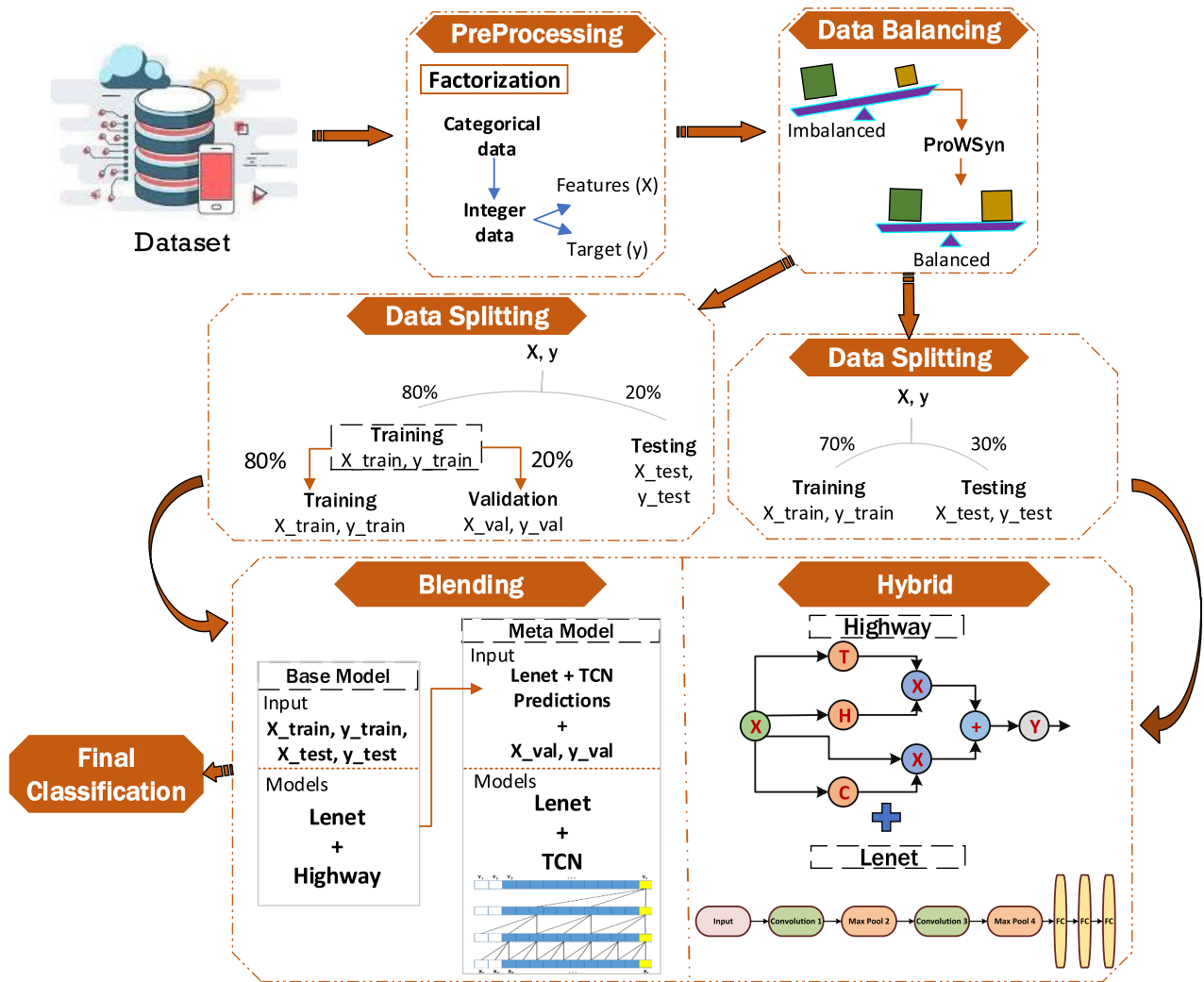


FIGURE 2. Proposed system models' diagram for diabetes prediction.

a Wi-Fi connection with a smartphone were used to gather information on blood glucose levels, body temperatures, and the amount of steps patients choose to take. The texts were then delivered to the attention-based deep GRU model. However, reliability and accuracy of the sensors deployed in the proposed work were considered the critical components of the proposed system's efficacy. Relying on Wi-Fi connectivity to transfer data could present problems, particularly in places with weak network coverage. In [30], the authors implied a robust ensemble-based approach for the prediction of diabetes mellitus called it the super learner model. The model was formed by four base models: LR, DT, RF and GradBoost, whose predictions were combined and passed to the meta model, i.e., support vector machine for final predictions. The authors adopted Chi-square for feature selection and GridSearch for self-hyper-parameter tuning purpose. The model outperformed the base models on three different datasets, i.e., diabetes 130-US hospitals, PIMA and early-stage diabetes risk prediction dataset. The

procedure for gathering data for the experiment uses two benchmark datasets in [31]. Firstly, the CNN was used to extract the deep selected features. The weighted entropy of deep features was extracted, where weight was adjusted using the Modified Escaping Energy-based Harris Hawks Optimization. The modified fuzzy classifier was used to process these features and classify high- and low-level glucose using the glucose level classification technique. Afterwards, the glucose prediction was performed using RNN and LSTM hybrid model, named R-LSTM. The model performed better than SVM, DNN, RNN and LSTM models. The authors in [32] suggested a method that end users could use to determine whether diabetes is present in patients or not based on various health indicators. For this system, the authors used Artificial Neural Networks (ANN) to create the prediction classifier. Because feature extraction is carried out automatically during the ANN model's training process, ANN is preferred over other approaches. This leads to a considerable reduction in training time, which becomes

especially noticeable when training with larger datasets. When compared to similar works using feature extraction, the model's classification accuracy of 0.8079 shows that this reduction in training time has little to no effect on the classification performance. An intelligent algorithm was created in [29] to determine whether such an exceeding limit has been required in the parameter, which may or may not involve urgency. The authors proposed an architecture using attention-based GRUs to construct this intelligent algorithm. A compact, portable device has been developed to measure body temperature. For individuals with diabetes, blood glucose levels are necessary to verify that the system is functioning properly. They offer the efficacy of different ML classifiers by differentiating between cases that were correctly and incorrectly classified in relation to the total number of instances. Performance was assessed using the following attributes: F1-score, recall, error rate, accuracy, and precision.

In [33], the authors present a self-explanatory ML interface for diabetes patients' diagnosis. Based on the publicly available diabetes dataset, they offer four classification models: DT, K-nearest Neighbor (KNN), Support Vector Classification (SVC), and Extreme Gradient Boosting (XGB). They used the ML interpretation technique called SHAP to clarify the inner workings of these models. The accuracy of the XGB model is a bit higher than that of the other models. They examined the XGB model using SHAP, offering detailed explanations of the logic underlying its predictions at a more refined scale. They then combined SHAP's local explanations with the XGB model to create an interface that can be used to predict a patient's risk of diabetes.

In [34], a MLP algorithm based on a DNN was utilized to perform early diabetes' prediction. Within a fully automated DL framework, hyperparameters were adjusted to maximize the efficiency of data preprocessing, prediction, and classification. The dataset used was Mansoura University Children's Hospital Diabetes (MUCHD), which included 18 important features and 548 patients. The system's consistency was confirmed by statistical measurements and cross-validation methods.

Particle Swarm Optimization (PSO) was used by the authors in [35] in order to perform feature selection, which was then followed by performance comparison. After that, the effectiveness of various ML techniques was compared using three medical datasets. The results of the experiments showed that DT, RF, and Naïve Bayes provided the best accuracy with the lowest mistake rate.

In summary, while prior research has used a variety of approaches and datasets to make substantial advancements in the prediction of diabetes, our work differs in a few important ways. Our novel methods involve combining three distinct DL architectures, highway networks, LeNet, and TCN, in order to improve proposed models' accuracy and early detection capabilities. In addition, we leverage the DPD and apply the ProWSyn to tackle important issues like unequal

Algorithm 1 ProWSyn: Proximity-Weighted Synthetic Oversampling

Require: Training data samples D_{tr} with m samples $\{(x_i, y_i)\}$, where x_i is an instance in n dimensional feature space X , and $y_i \in \{-1, 1\}$ is the class identity level associated with x_i .

- 1: Define S_{maj} and S_{min} to be the majority and minority class sets respectively, with m_l and m_s as the number of majority and minority class samples, where $m_s \leq m_l$ and $m_s + m_l = m$.
 - 2: **procedure** ProWS(D_{tr}, β)
 - 3: Calculate the number of synthetic samples to be generated for the minority class: $G = (m_l - m_s) \times \beta$
 - 4: Initialize $P = S_{min}$
 - 5: **for** $i = 1$ to $L - 1$ **do**
 - 6: From each majority sample y , find the nearest K minority samples in P according to Euclidean distance. Let the set of these K samples be $N_K(y)$.
 - 7: Form partition P_i as the union of all $N_K(y)$:
 - 8: $P_i = \bigcup_{y \in S_{maj}} N_K(y)$
 - 9: Set proximity level of each minority sample x in partition P_i to be i : $P_{Lx} = i, \forall x \in P_i$
 - 10: Remove selected minority samples from P : $P = P - P_i$
 - 11: **end for**
 - 12: Form partition P_L with the remaining unpartitioned samples in P .
 - 13: Set proximity level of each x in P_L to be L : $P_{Lx} = L, \forall x \in P_L$
 - 14: **for each** x **do**
 - 15: Calculate a weight w_x from its proximity level P_{Lx} : $w_x = \exp(-\theta \cdot (P_{Lx} - 1))$
 - 16: **end for**
 - 17: Normalize w_x according to $w'_x = w_x / \sum_{z \in S_{min}} w_z$, so that w'_x is a density distribution ($\sum_{z \in S_{min}} w'_z = 1$)
 - 18: **for each** x **do**
 - 19: Calculate the number of synthetic samples g_x that need to be generated for x : $g_x = w'_x \times G$
 - 20: **end for**
 - 21: Find the clusters of minority set, S_{min}
 - 22: Initialize set, $S_{omin} = S_{min}$
 - 23: **for each** x **do**
 - 24: **for** $j = 1$ to g_x **do**
 - 25: Randomly select one minority sample y from x 's cluster (as found in step 9).
 - 26: Generate a synthetic sample s according to: $s = x + \alpha \times (y - x)$, where α is a random number in the range $[0, 1]$.
 - 27: Add s to S_{omin} : $S_{omin} = S_{omin} \cup \{s\}$
 - 28: **end for**
 - 29: **end for**
 - 30: **Output:** Oversampled minority data set, S_{omin}
 - 31: **end procedure**
-

class distribution, and to ensure the stability and performance of our models. Also we have implemented 10-FCV to validate our models' results and to know about features' contributions in diabetes prediction. Our study advances the state-of-the-art in diabetes prediction by introducing these novel methodologies and demonstrating their superior performance. Also by providing insightful information and practical solutions for enhancing healthcare outcomes. Also, it can be seen from Table 2 how our proposed model is more novel than prior studies.

Algorithm 2 Flow of Information in Highway Network

Require: Input x, W_T, W_C, b_T, b_C

Ensure: Model output

```

1: Initialize  $T, C, y, H, carriedinput, output$ 
2: for each element in  $x$  do
3:   // Compute transition gate using sigmoid:
4:    $T \leftarrow \sigma(W_T \cdot \text{element} + b_T)$ 
5:   // Compute complementary carry gate:
6:    $C \leftarrow 1 - T$ 
7:   // Transform input with  $W_T$  to get transformed
   information:
8:    $y \leftarrow \text{activation}(W_T \cdot \text{element} + b_T)$ 
9:   // Generate transformed information:
10:   $H \leftarrow T \cdot y$ 
11:  // Generate carried information:
12:   $carriedinput \leftarrow C \cdot \text{element}$ 
13:  // Combine transformed and carried information:
14:   $output \leftarrow H + carriedinput$ 
15:  // Apply sigmoid activation to the combined output:
16:  Apply sigmoid layer to  $output$ 
17: end for
18: return  $output$ 
  
```

III. PROPOSED SYSTEM MODEL

In the proposed work, we have implemented two ensemble learning methods, hybrid and blending, for diabetes prediction. Hi-Le hybrid model is designed by concatenating the prediction of two DL models, Highway and LeNet. While HiTCL blending model is designed using three DL models TCN, LeNet and Highway. The prediction and validation data of the base layer models (TCN and LeNet) was fed to the meta layer model (Highway) for final diabetes predictions, as shown in Figure 2. For these models' implementation, DPD dataset is used which is highly imbalanced. So, ProWSyn technique is used to balance the dataset.

A. DATASET

In this study, we have utilized the DPD to predict the presence of diabetes in healthy people. Several Electronic Health Record (EHR) sources provided DPD data, which were aggregated to create the DPD. Along with demographic information, EHRs provide an enormous amount

of patient data. Furthermore, EHRs generate data that eventually becomes patient health information. The dataset is publicly available on the Kaggle repository. The dataset has 100000 occurrences and 9 characteristics [36]. The dataset contains statistical and medical information on diabetes patients in addition to patient-specific information. The characteristics of the dataset include age, gender, blood sugar level, smoking history, heart disease, hypertension, Body Mass Index (BMI), and diabetes status indicated as the objective class. During data preparation, the dataset's original categorical characteristics are converted to integers. Furthermore, there is a significant imbalance in the dataset and a wide range in the feature values. The imbalance ratio between class 0 and class 1 is 91.5% and 8.5%.

B. DATA PRE-PROCESSING

This section contains pre-processing on the dataset. The data contains categorical features and highly imbalanced ratios, which are dealt with before deep models' implementation.

Algorithm 3 Flow of Information of LeNet-5 Model in Diabetes Prediction

Require: Input data

Ensure: Network output

```

1: Initialize LeNet-5 layers and parameters
2:  $input \leftarrow$  Input data
3: Stage 1: Extract Features
4:  $feature\_map\_C1 \leftarrow$  ConvolutionLayer( $input, weights\_C1,$ 
    $biases\_C1$ )
5:  $feature\_map\_S2 \leftarrow$  MaxPooling( $feature\_map\_C1,$ 
    $pool\_size\_S2$ )
6: Stage 2: Deepen Features
7:  $feature\_map\_C3 \leftarrow$  ConvolutionLayer( $feature\_map\_S2,$ 
    $weights\_C3, biases\_C3$ )
8:  $feature\_map\_S4 \leftarrow$  MaxPooling( $feature\_map\_C3,$ 
    $pool\_size\_S4$ )
9: Stage 3: Refine Features
10:  $feature\_map\_C5 \leftarrow$  ConvolutionLayer( $feature\_map\_S4,$ 
    $weights\_C5, biases\_C5$ )
11: Stage 4: Transform for Classification
12:  $flattened\_vector \leftarrow$  Flatten( $feature\_map\_C5$ )
13:  $fully\_connected\_output \leftarrow$ 
   FullyConnectedLayer( $flattened\_vector, weights\_F6,$ 
    $biases\_F6$ )
14: Output Layer 5: Activation for Prediction
15:  $Network\_output \leftarrow$ 
   ActivationFunction( $fully\_connected\_output$ )  $\triangleright$  Apply
   activation function
16: return  $Network\_output$   $\triangleright$  Return network output
  
```

1) FEATURE ENCODING

In this dataset, there are two features, gender and smoking_history, that contain categorical values, and most of the DL models do not work with categorical features. There are two ways to implement deep models: either to drop that column(s) or to encode them into an integer or floating type. In this study, we have implemented the *factorize* method for encoding [37]. It transforms categorical data into numerical

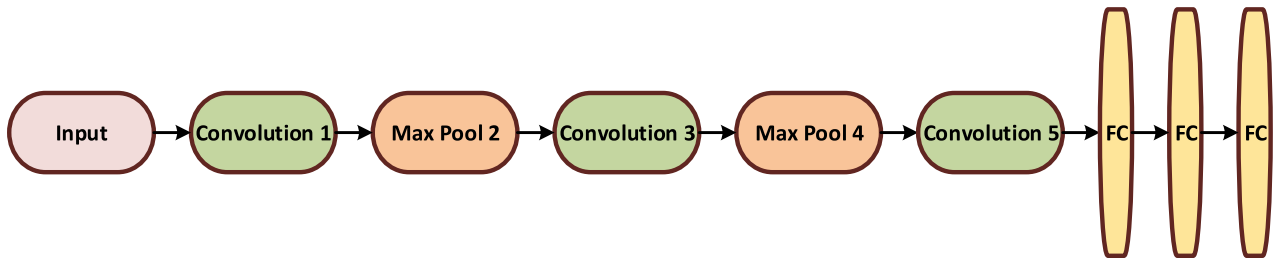


FIGURE 3. LeNet architecture.

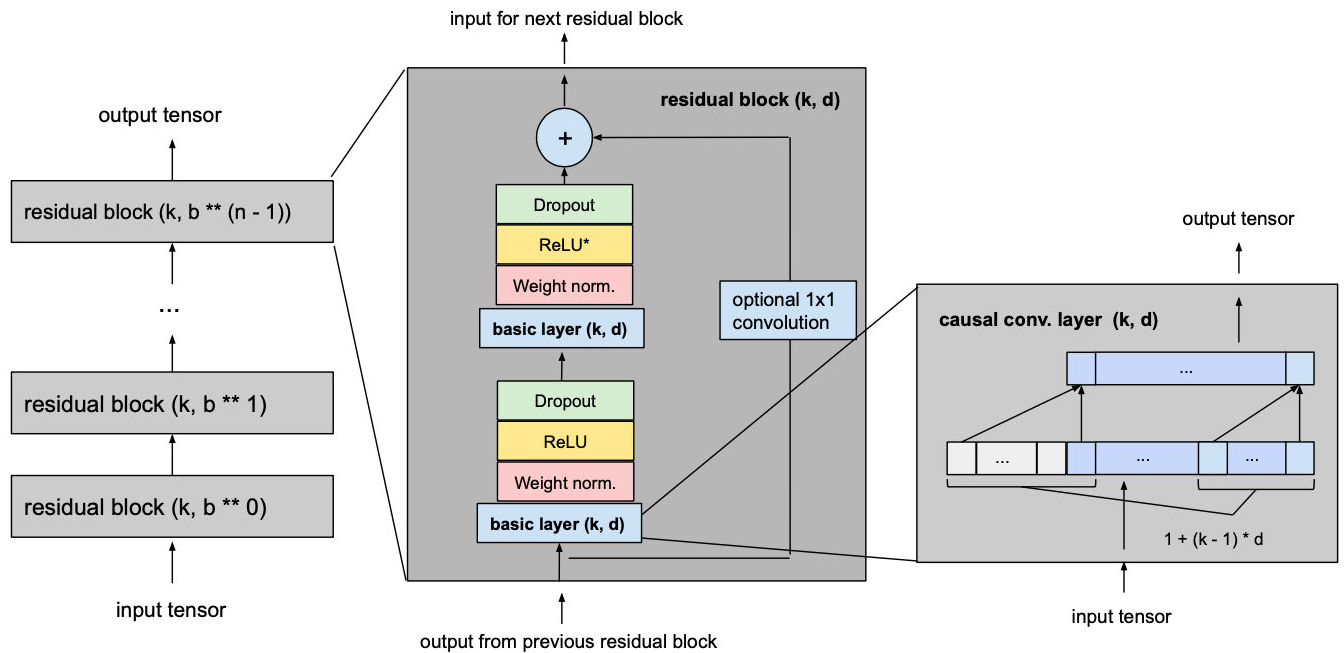


FIGURE 4. Temporal convolutional network architecture.

format by allocating a unique integer to each unique value in a category column. The encoded values are returned as an array of integers by the procedure.

2) DATA BALANCING

The distribution of data among classes is a significant component that influences the performance and capacity for generalization of a model. When one class contains more data instances than the other, there is an imbalance in the data. A minority class has the fewest data instances, whereas a majority class has the greatest number of data instances. Because models get over-fit on data from the dominant class when there is a class imbalance, the findings are skewed and cannot be applied to whole datasets.

As human life is in danger, so in the healthcare industry, accurate predictions are crucial, and this can only be achieved by training the model uniformly across all output classes and on challenging data. The DPD dataset is highly imbalanced, class 0, which indicates no diabetes, contains

91500 instances, and class 1, which represents diabetic patients, has 8500 data points. There are two possible solutions for this imbalanced dataset: either under-sampling or oversampling the original data to get accurate and generalizable results from models. If we have applied under-sampling, it will reduce the non-diabetic individual class instances to diabetic patients. By doing this, we will lose a lot of information and data patterns that may be important in diabetes detection and prediction. So, we have avoided this process. The second solution is oversampling, in which diabetic patients data is synthetically generated to get balanced data. For this synthetic generation, a lot of techniques and algorithms have been implemented.

Proximity-Weighted Synthetic Oversampling (ProWSyn) Technique: In this paper, we have implemented the ProWSyn, an oversampling technique, that is inspired by the weight generation process of Cluster Based Synthetic Oversampling (CBSO) [38], and then weights are used for synthetic sample generation. The ProWSyn technique's primary goal is to give

minority-class samples weights based on their significance. ProWSyn operates across two stages. Minority data is divided by the number of divisions in the first phase according to how far they are from the decision boundary. Next, it gives each division a proximity level; the partition closest to the decision border is given the lowest proximity level, and the level rises as the distance from the decision boundary increases. As samples nearest to the decision boundary are important and hard to learn, they are important for synthetic sample generation as compared to the samples that are far away from the decision boundary [39]. In phase two, ProWSyn generates additional synthetic minority class samples in lower proximity locations, or places that are extremely near to the decision boundary, using the proximity information.

Algorithm 4 Temporal Convolutional Network (TCN) for Diabetes Prediction

Require: Input X , TCN parameters

Ensure: Model output

```

1: Initialize TCN layers and parameters
2: Initialize weights and biases for each layer
3:  $input \leftarrow X$ 
4: for  $i$  in 1 to  $tcn\_layers$  do
5:   Temporal Convolutional Layer  $i$ :
6:    $conv\_output_i \leftarrow \text{Convolution}(input, weights\_conv_i, biases\_conv_i)$ 
7:    $input \leftarrow conv\_output_i$   $\triangleright$  Update input for the next layer
8: end for
9: Flatten:
10:  $flattened\_vector \leftarrow \text{Flatten}(conv\_output_{tcn\_layers})$ 
11: Fully Connected Layer:
12:  $fc\_output \leftarrow \text{FullyConnect}(flattened\_vector, weights\_fc, biases\_fc)$ 
13: Output Layer:
14:  $Network\_output \leftarrow \text{Activation}(fc\_output)$   $\triangleright$  Apply activation function
15: return  $network\_output$   $\triangleright$  Return network output
```

Using the Euclidean distance, ProWSyn determines which K minority class samples are closest to each majority sample. The first partition, P1 (of proximity level 1), is the collection of all these minority class samples, or the level of samples closest to the border. The next step is to measure the distance between each majority and the K minority samples. These samples, when combined, comprise the second partition (P2 of proximity level 2). The algorithm 1 [39] provides thorough and detailed information.

C. HIGHWAY NETWORKS

The highway network is a unique kind of Neural Network (NN) with gating mechanisms and a route that allows data to flow across several layers without degrading. These routes are referred to as information highways, and these networks are called highway networks. A Highway Network, which explicitly models information flow through a controlled

mechanism, enhances the tabular data modeling process. Highway routes function as skip connections, allowing data to flow directly across networks. In order to address the problem of vanishing gradients in DNN, highway networks were built in 2015 [40]. As the number of layers increases, typical NN optimization worsens. However, due to their pathway idea, Highway Networks are unaffected by the rise in layer count. A simple NN is made up of L layers. Its l th layer ($l \in \{1, 2, \dots, L\}$) produces its output y_L by executing a nonlinear transform H (parameterized by W_H, l) on its input x_1 , where the model's output is y_L and its input is x_1 [40]. The biases and layer indexes are disregarded in the interest of simplicity.

$$y = H(x, W_H) \quad (1)$$

where a non-linear transform function might come after the linear transform H . Two nonlinear transformations are used in Highway Networks to regulate the information flow in each tier.

- Transform Gate ($T(x, W_T)$)
- Carry Gate ($C(x, W_C)$)

Diabetes input data is fed to the Highway Network, where each feature contains some critical information. Next, a transform gate determines which parts of the input should be changed and which ones should go straight to the next layer by applying a sigmoid non-linear transform. By using the sigmoid non-linear activation function, the transform gate is computed. The network constantly modifies these gate settings to effectively prioritize input based on their significance for diabetes prediction. We also delve into more details about how the transform gate and carry gate control the flow of information inside each layer, highlighting how crucial they are to the process of selecting data for diabetes prediction.

1) TRANSFORM GATE

In a highway network, calculating the transform gate is the initial step. The transform gate determines how much of the input should go straight to the next layer and how much should be transformed. The transform gate is calculated using a non-linear activation function called the sigmoid function. The formula for transform gate is given in equation 2.

$$T(x) = \sigma(W_T^T x + b_T) \quad (2)$$

where W_T is the weight matrix and b_T is the bias vector for the transform gates [40].

2) CARRY GATE

The carry gate determines how much of the input should be directly carried forward without any transformation. Highway paths are actually implemented by a carry gate. For simplicity, the carry gate is defined as the inverse of the transform gate, and is given in equation 3 [40].

$$C(x, W_C) = 1 - (T(x, W_T)) \quad (3)$$

TABLE 3. Training results of HiTCL model.

Epochs	Models	Accuracy	F1-Score	Recall	Precision	E.Time(s)
5	LeNet	0.89	0.90	0.96	0.86	41
	TCN	0.88	0.89	0.90	0.87	135
	Highway	0.89	0.90	0.90	0.90	42
	HiTCL	0.92	0.92	0.91	0.93	11
10	LeNet	0.89	0.89	0.90	0.87	72
	TCN	0.91	0.91	0.88	0.95	226
	Highway	0.90	0.91	0.90	0.90	83
	HiTCL	0.94	0.94	0.91	0.95	24
20	LeNet	0.78	0.79	0.82	0.76	153
	TCN	0.92	0.92	0.93	0.92	537
	Highway	0.86	0.87	0.86	0.88	124
	HiTCL	0.94	0.93	0.93	0.93	32

TABLE 4. Testing results of HiTCL model.

Epochs	Models	Accuracy	F1-Score	Recall	Precision	E.Time(s)
5	LeNet	0.90	0.90	0.96	0.85	25
	TCN	0.88	0.89	0.90	0.87	57
	Highway	0.90	0.90	0.90	0.90	23
	HiTCL	0.92	0.92	0.91	0.93	7
10	LeNet	0.89	0.89	0.91	0.87	25
	TCN	0.91	0.91	0.88	0.95	57
	Highway	0.90	0.91	0.90	0.90	23
	HiTCL	0.94	0.94	0.91	0.96	7
20	LeNet	0.78	0.79	0.82	0.76	25
	TCN	0.92	0.92	0.93	0.92	57
	Highway	0.86	0.86	0.86	0.86	23
	HiTCL	0.94	0.93	0.92	0.94	7

3) TRANSFORMATION LAYER

Highway networks also play a crucial role in transformation. Equation 1 illustrates how the transformation function is composed of completely linked layers followed by non-linear transformations. The number of layers and non-linear transforms can change depending on the circumstances, however we have included four dense layers with sigmoid non-linear transforms. Since the data is binary categorized, we have chosen to utilize the Adam optimizer with binary cross entropy as the loss function. One can use equation 4 to produce output Y .

$$Y = H(x, W_H).T(x, W_T) + x.C(x, W_C) \quad (4)$$

As we know, the carry gate is the inverse of the transform gate. So the resultant will be acquired using equation 5.

$$Y = H(x, W_H).T(x, W_T) + x.(1 - (T(x, W_T))) \quad (5)$$

In order to give a better understanding of how Highway Networks work, we incorporate the explanation of the information flow in Algorithm 2.

D. LeNet-5

LeNet-5 is a sequential and simpler architecture than other CNN designs, making it easy to learn and apply. To anticipate the presence of diabetes, we leverage LeNet's ease of use and capacity for pattern recognition. Three convolution layers, two max pooling layers, and two dense layers make up LeNet-5's seven layers, as shown in Figure 3 [42]. But LeNet-5 gets its name from the fact that it has just five learnable layers. The first layer uses five kernels and six

neurons to conduct 1D convolution on the input of one-Dimensional (1D) diabetes data. This is followed by a max pooling layer. While the max pooling layer is used to down sample feature maps by using data point with highest feature, the convolution layer is crucial for uncovering hidden patterns and intricate correlations in the data. The number of consecutive characteristics that may be applied simultaneously to convolution is indicated by the kernel size. The same kernel size and 16 neurons are employed in the second convolution layer. A layer of max pooling follows it as well. Two dense layers with 120 and 84 neurons are applied after the third convolution layer, which uses 1×1 convolution with one neuron. The sigmoid activation function is applied at the top layer. However, the tan hyperbolic activation function is applied in the remaining 5 levels. The information flow in LeNet-5 is described in Algorithm 3, which presents how the model identifies and improves features for diabetes prediction.

E. TEMPORAL CONVOLUTIONAL NETWORK (TCN)

TCN use a family of convolutional architectures to process sequential data. The TCN model ensures causal constraints such that no information from future time steps can be used for prediction; only information from current and previous time steps can be used for model prediction, which is not ensured in other sequence models. TCN uses the architecture of a 1-Dimensional Fully Connected Network (1DFCN) [43]. To ensure the same sequence length in all layers, 0-padding is used so that each hidden layer has the same sequence length as the input length. In simple convolution, the model

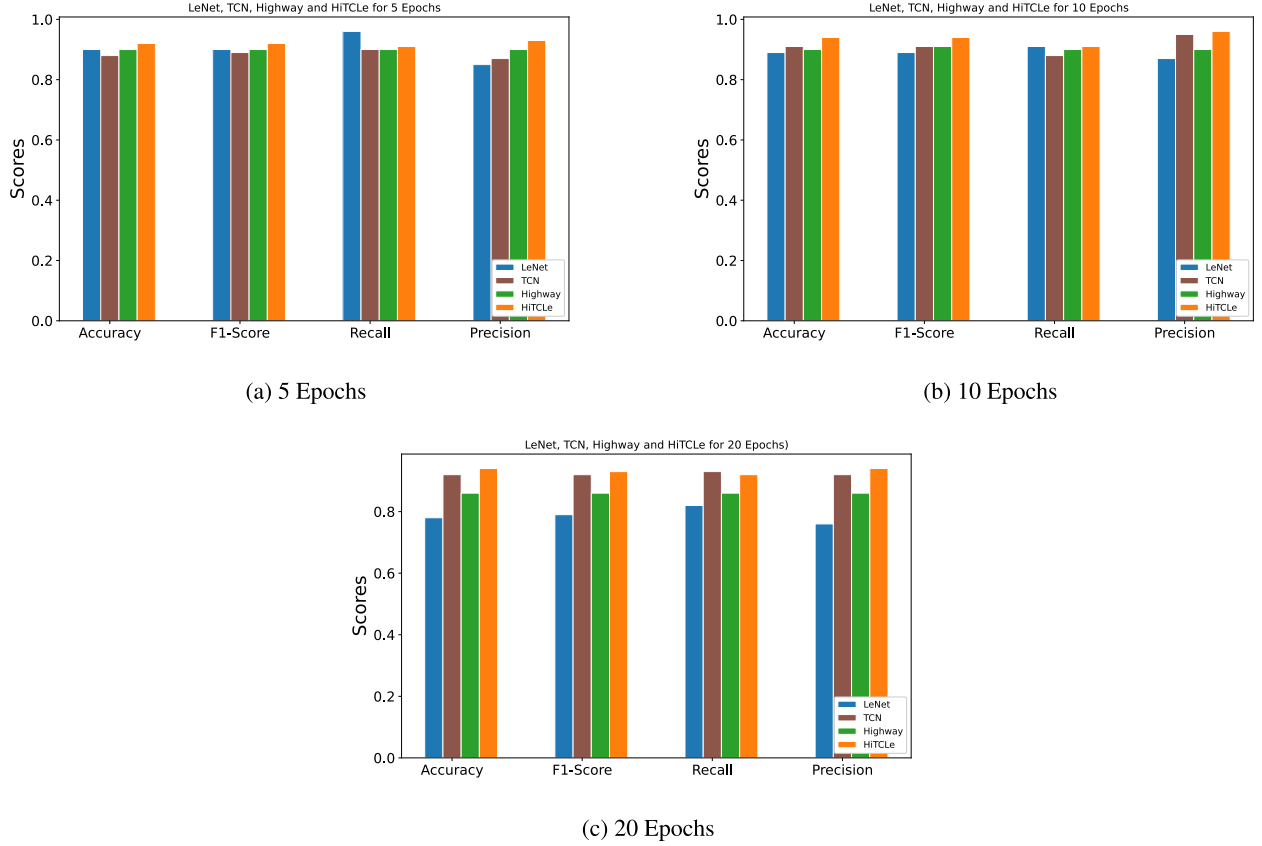


FIGURE 5. Comparison between LeNet, TCN, highway, and HiTCLe.

looks back to consecutive time steps, which results in a small receptive field size. To overcome this issue, dilated convolution is used in the TCN model, which aids the model to look back up to $(k-1)d$ time steps, and d grows exponentially with the depth of the model

$$d = O(2^i) \quad (6)$$

where i is the level of the network and it starts from 0 [43]. We can calculate number of layers (n) required to cover complete temporal coverage.

$$n = \left\lceil \log_b \left(\frac{(l-1) \cdot (b-1)}{(k-1)} + 1 \right) \right\rceil \quad (7)$$

where k is the size of kernel, b is dilation base, $k \geq b$ and l is the input length [43].

The concept of causal and dilated convolution is taken from wavenet model [44]. The basic building block in TCN is residual block which is made up of two layers with same dilation rate and a residual connection instead of an ordinary 1D casual convolution layer. A residual block consists of 2 layers of casual convolution, weight normalization, ReLU activation function and dropout. Also there is an optional 1×1 convolution in case if number of channels differ from number of output channel for ensuring element wise addition of residual block and input layer like ResNet model [45]. Figure 4 shows the complete structure and working of TCN model. The full functionality of TCN is given in Algorithm 4,

which shows how the model extracts temporal characteristics for diabetes prediction.

F. HiTCLe

HiTCLe is a blend of the highway network, TCN, and LeNet models. Blending is an ensemble learning method that best combines different DL models' predictions to give more accurate and reliable predictions because a single model can give us wrong results, but a combination of different models can not give us wrong predictions. In blending, predictions are made up of holdout (validation) data only. The validation set and predictions of base models are used to build a meta model, which is then tested on the test set [46]. Steps to build a blending model are given below:

- 1) Firstly, the dataset is divided into train and test sets.
- 2) Then, the train set is divided into validation and train sets.
- 3) Base models are trained on training data and tested on validation data.
- 4) Validation sets and base model predictions are used as features to train the meta model, and test data is used for the final prediction.

In HiTCLe, LeNet and TCN are used as base layer models, and Highway is used as a meta layer model. A combination of these models performs best in diabetes prediction due to the diversity of the models. LeNet typically consists of

TABLE 5. Hyperparameters of highway, LeNet, TCN and HiTCLe models.

Common Parameters:	
Optimizer	Adam
Loss Function	Binary Cross_entropy
Metrics	Accuracy
Number of Epoch	5-20
Batch Size	64
Activation Function on Output Layer	Sigmoid
Highway Network:	
Hidden Layers	4
Activation Function in every Hidden layer	ReLU
Number of Neurons	32-64
LeNet	
kernel Size	3
Number of Neurons	1-64
Activation Function	tanh
TCN:	
Filter size	4-32
Kernel size	3
Dilation rate	1-8
Spatial dropout rate	0.4
Activation	ReLU
HiTCLe:	
Hidden Layers	4
Activation Function in every Hidden layer	ReLU
Number of Neurons	32-64

convolution layers followed by subsampling layers, which capture hierarchical spatial dependencies within feature space. TCN uses 1D convolution and is capable of capturing temporal dependencies, which is useful for time-dependent features in the diabetes dataset. Highway has the potential to control information flow; it can selectively learn and pass relevant information and adapt various patterns present in the dataset.

G. Hi-Le MODEL

The Hi-Le model is a hybrid of the Highway and LeNet models. In the Hi-Le model, data after balancing is passed to both individual models, Highway and LeNet, as input. Both models work independently and in parallel. At the end, the predictions of both individual models are concatenated to generate a hybrid model named Hi-Le. The combination of Highway and LeNet performs best because they both are diverse. The LeNet model is a convolutional type that is best at capturing the spatial relationships of features in data, whereas the highway model controls the flow of information between layers. By demonstrating the advantages of integrating Highway's adaptive information processing with LeNet's spatial hierarchies, we can demonstrate how the Hi-Le model outperforms other diabetes prediction models.

IV. SIMULATION RESULTS AND DISCUSSION

The simulations are conducted on a cloud-based platform called Google Colab, which is well known in DL research. With Colab, we have utilized Python 3.x, TensorFlow 2.x, and PyTorch 1.x. We have also accessed NVIDIA Tesla

K80 GPUs for models for early and accurate training and detection. The experiments are conducted on DPD, and the results demonstrate that hybrid and blending models give more accurate predictions about whether an individual is suffering from diabetes or not. The proposed models' performance is evaluated using accuracy, F1-Score, recall, and precision evaluation metrics. Both Hi-Le and HiTCLe models are trained and tested on oversampled data as our models are exposed to a wider variety of instances when they are trained and tested only on augmented data, which improves their generalizing abilities. Our models are able to learn more resilient and flexible patterns thanks to the addition of augmented data, which presents variations and scenarios that may not be sufficiently represented in the original dataset alone. Also, ProWSyn augmented data creates synthetic instances of minority class samples in order to solve the data imbalance issue.

A. HiTCLe

Experimental results show that the HiTCLe blending model performs the best as compared to its individual models, as shown in Table 4. The HiTCLe model achieves the highest accuracy of 94%, F1-Score of 94%, recall of 92% and precision of 96%. The combination of these models performs best on the DPD dataset, as LeNet is best at capturing spatial and hierarchical relationships between features of data and TCN is best at capturing temporal and time-dependent features of data that may be important in diabetes prediction. The highway is well-known for its ability to learn which information to bypass and which to process through the network. This is beneficial for diabetes prediction. The highway model is fed with predictions from the LeNet and TCN models along with validation data, so it is easy for the highway to learn short- and long-term dependencies. Highways have skip connections, which help model the flow of information, prevent vanishing gradient problems, and improve gradient flow during training. Graphical representations of these results are also shown in Figure 5 on 5, 10, and 20 epochs. The complete architecture of individual models and HiTCLe models' hyperparameters and their architectures are given in Tables 5 and 6, respectively. Each feature in a tabular data analysis corresponds to a particular attribute or quality of the data. We ensure that no data is lost during the pooling process by employing "Max Pooling (pool_size=1)" in LeNet, thereby maintaining the integrity of every feature. This is especially important in tabular data settings, where capturing complex patterns and relationships within the data requires maintaining the minute details of individual features.

1) ABLATION STUDY

The scientific and experimental technique known as "ablation study" is used to determine the effects of specific system elements by "ablating" or removing them from the system and then observing how their absence impacts the system's overall performance. Ablation studies are frequently

TABLE 6. Highway, LeNet, TCN and HiTCLC models' architectures.

Model	Architecture
Highway Network	input = Input(shape=(X_train.shape[1],)) Transformation layer: Dense(neurons: 8, activation: sigmoid) Dense(neurons: 64, activation=ReLU) Dense(neurons: 64, activation=ReLU) Dense(neurons: 32, activation=ReLU) Dense(neurons: 32, activation=ReLU) Dense(neurons: 2, activation=Sigmoid)
LeNet	input = Input(shape=(X_train.shape[1],)) Conv1D (neurons: 6, kernel_size: 3, activation: tanh) Max Pooling (pool_size=2) Conv1D (neurons:32, kernel_size: 3, activation: tanh) Max Pooling (pool_size=1) flatten() Dense (neurons:16, activation=tanh) Dense (64, activation=tanh) Dense (1, activation=sigmoid)
TCN	def tcn_layer(inputs, filters, kernel_size, dilation_rate) Conv1D(filters=filters, kernel_size= kernel_size, dilation_rate= dilation_rate, padding= causal) LayerNormalization Activation= relu SpatialDropout1D(0.4) tcn_layer(filters=32, kernel_size=3, dilation_rate=1) tcn_layer(filters=16, kernel_size=3, dilation_rate=2) tcn_layer(filters=8, kernel_size=3, dilation_rate=4) tcn_layer(filters=4, kernel_size=3, dilation_rate=8) Dense(neurons=1, activation='sigmoid') tcn_input=(X_train.shape[1],1) tcn_model = tcn_layer(tcn_input, filters, kernel_size, dilation_rate)
HiTCLC Model	Transformation layer: Dense(neurons: 8, activation: sigmoid) Dense(neurons: 64, activation=ReLU) Dense(neurons: 64, activation=ReLU) Dense(neurons: 32, activation=ReLU) Dense(neurons: 32, activation=ReLU) Dense(neurons: 1, activation=Sigmoid)

TABLE 7. HiTCLC Model results with SGD optimizer.

Models	Accuracy	F1-Score	Recall	Precision	E.T(sec)
LeNet	0.73	0.72	0.74	0.69	83
TCN	0.74	0.67	0.93	0.52	204
Highway	0.80	0.83	0.80	0.80	83
HiTCLC	0.81	0.80	0.81	0.79	27

TABLE 8. HiTCLC model results with batch size 32.

Models	Accuracy	F1-Score	Recall	Precision	E.T(sec)
LeNet	0.90	0.90	0.87	0.94	83
TCN	0.91	0.91	0.93	0.89	264
Highway	0.88	0.89	0.88	0.88	143
HiTCLC	0.91	0.91	0.90	0.92	49

employed in the DL to examine the relative contributions of various elements, attributes, or modules. The results in Tables 3 and 4 shows the results comparison between individual models and the HiTCLC model, as well as the different epoch sizes to check model prediction. Table 7 shows results by changing the optimizer from Adam to SGD, and results show it does not work well on individual models and also on HiTCLC model. Similarly, Table 8 shows the results obtained with batch size 32. The results shows that HiTCLC model did not work well with this batch size, and its execution time is high.

2) SHAPLEY ADDITIVE EXPLANATIONS (SHAP)

A framework called SHapley Additive exPlanations (SHAP) is used to interpret intricate DL models by allocating a contribution to each feature's final output. The fundamental principle of SHAP values derives from cooperative game theory, more especially from Shapley values, which seek to equitably divide the "payoff" among participating players. With the help of SHAP values, one can assign a model's prediction to every single feature, gaining insight into the significance and influence of each feature on the model's output. DL models in particular, which frequently serve as complex black boxes, present a challenge to interpretability that SHAP values aim to address. These models can have a large number of parameters and complex interactions, which makes it difficult to comprehend how decisions are made. The additive nature of SHAP values allows users to understand the behavior of the model at the feature level by dissecting the model's prediction into the contributions from individual features.

The capability of SHAP values to offer easily understandable explanations for predictions makes them significant when it comes to DL models. DL models must be interpretable in order to be used in sensitive fields where decisions affect people's lives, like criminal justice, healthcare, and finance. A thorough understanding of feature interactions and how they affect predictions is provided by SHAP values.

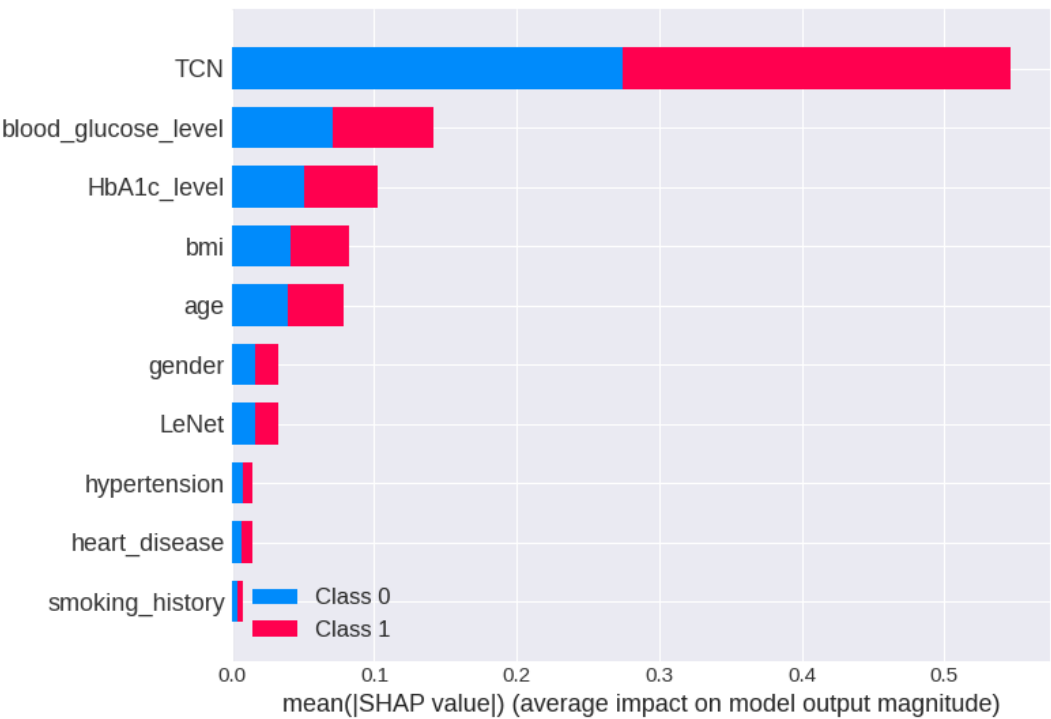


FIGURE 6. Summary plot for HiTCLe model.

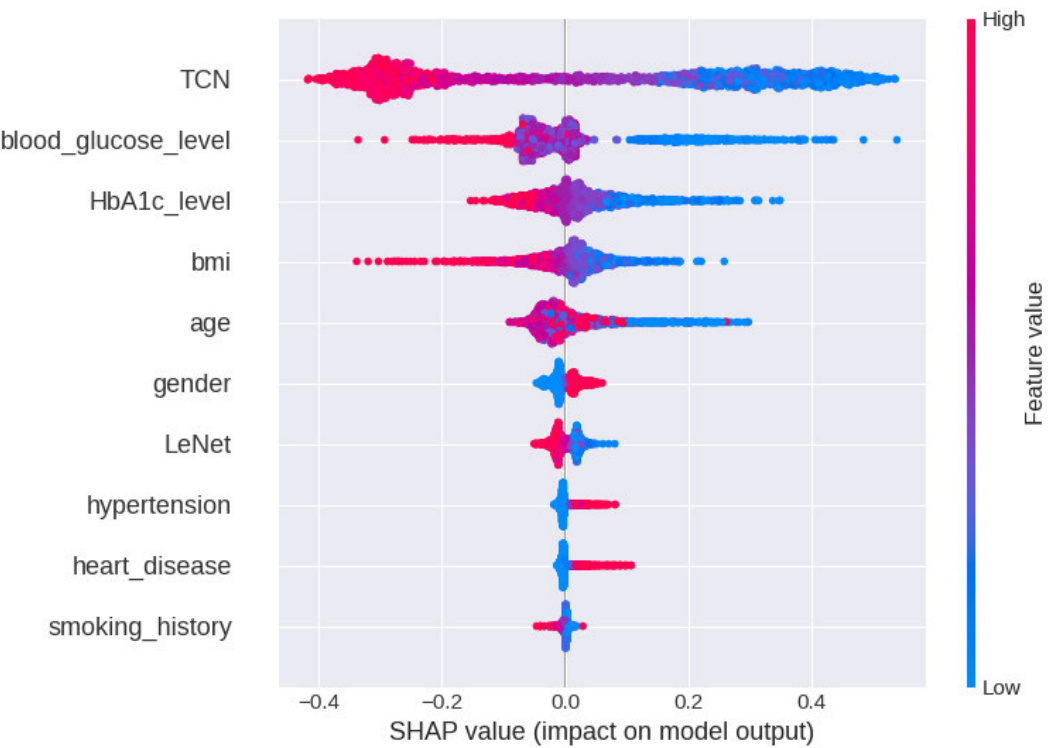


FIGURE 7. Beeswarm plot for HiTCLe model.

This information is useful for feature engineering, improving the model, debugging the model, and improving the comprehension of intricate relationships in the data. Enhancing interpretability is one way that SHAP values help achieve

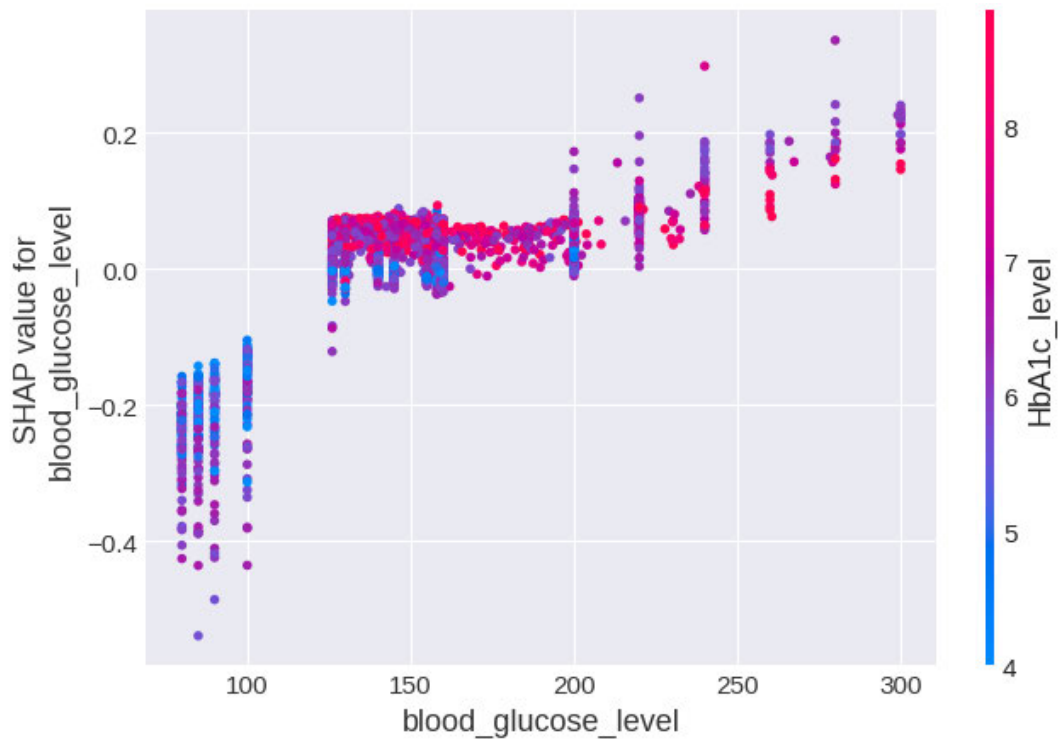


FIGURE 8. Dependence plot for HiTCLe model.

TABLE 9. Training results for highway, LeNet and Hi-Le model.

Epochs	Models	Accuracy	F1-Score	Precision	Recall	E.Time(s)
5	Highway	0.85	0.85	0.84	0.86	124
	LeNet	0.78	0.79	0.82	0.76	153
	Hi-Le	0.94	0.95	0.92	0.92	71
10	Highway	0.90	0.91	0.91	0.91	83
	LeNet	0.90	0.89	0.91	0.87	72
	Hi-Le	0.94	0.96	0.93	0.94	124
20	Highway	0.86	0.87	0.88	0.86	124
	LeNet	0.78	0.79	0.81	0.77	153
	Hi-Le	0.94	0.96	0.94	0.95	180

TABLE 10. Testing results for highway, LeNet and Hi-Le model for 5 epochs.

Models	Accuracy	F1-Score	Precision	Recall	E.Time(s)
Highway	0.86	0.86	0.86	0.86	37
LeNet	0.88	0.88	0.89	0.87	49
Hi-Le	0.94 , 0.93, 0.94, 0.94, 0.94, 0.94, 0.94, 0.94, 0.93, 0.94	0.94 , 0.93, 0.94, 0.94, 0.94, 0.94, 0.93, 0.94, 0.94, 0.94	0.95 , 0.96, 0.99, 0.95, 0.95, 0.94, 0.96, 0.95, 0.95, 0.95	0.92 , 0.88, 0.92, 0.93, 0.94, 0.92, 0.94, 0.92, 0.94, 0.93	71 , 84, 43, 39, 43, 72, 45, 46, 44, 84, 43, 55

TABLE 11. Testing results for highway, LeNet and Hi-Le model for 10 epochs.

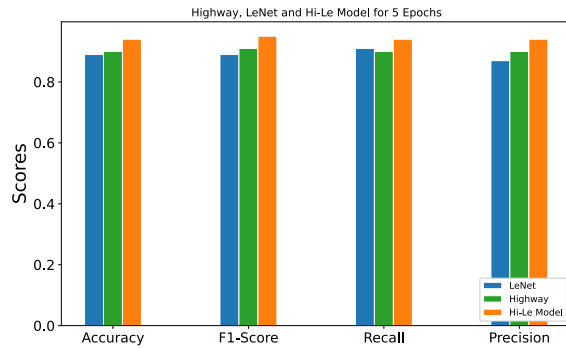
Models	Accuracy	F1-Score	Precision	Recall	E.Time(s)
Highway	0.90	0.91	0.90	0.90	83
LeNet	0.89	0.89	0.91	0.87	72
Hi-Le	0.94 , 0.94, 0.94, 0.95, 0.94, 0.94, 0.94, 0.94, 0.94, 0.94	0.96 , 0.94, 0.94, 0.95, 0.94, 0.94, 0.94, 0.94, 0.94, 0.94	0.93 , 0.93, 0.97, 0.96, 0.94, 0.95, 0.98, 0.94, 0.96, 0.95	0.94 , 0.95, 0.92, 0.93, 0.94, 0.94, 0.91, 0.94, 0.93, 0.93	124 , 84, 83, 84, 80, 89, 80, 85, 86, 147, 84, 90

the larger objective of developing reliable and responsible DL systems.

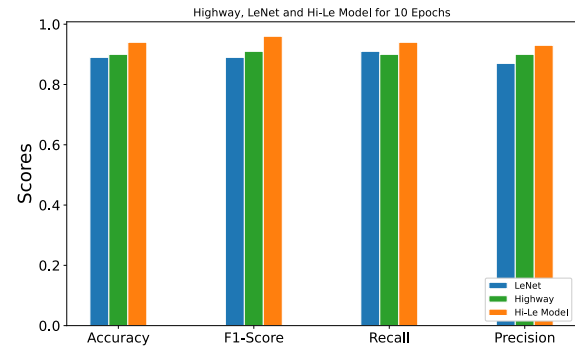
SHAP on HiTCLe Model: For the blending model HiTCLe, the SHAP summary plot provides insightful information

TABLE 12. Testing results for highway, LeNet and Hi-Le model for 20 epochs.

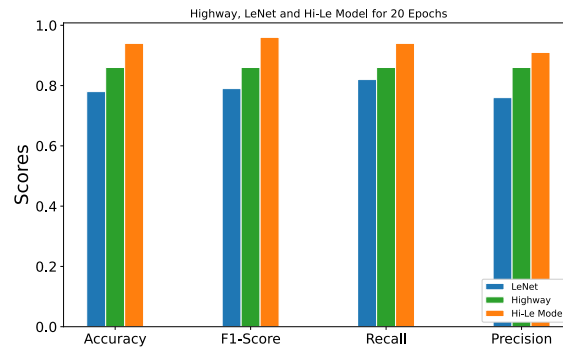
Models	Accuracy	F1-Score	Precision	Recall	E.Time(s)
Highway	0.86	0.86	0.86	0.86	124
LeNet	0.78	0.79	0.82	0.76	153
Hi-Le	0.94 , 0.95,	0.96 , 0.95,	0.94 , 0.96,	0.93 , 0.93,	208 , 160,
	0.95, 0.95,	0.95, 0.95,	0.94, 0.96,	0.95, 0.95,	150, 155,
	0.95, 0.95,	0.95, 0.95,	0.98, 0.97,	0.92, 0.93,	204, 204,
	0.95, 0.95,	0.95, 0.95,	0.97, 0.97,	0.93, 0.92,	205, 204,
	0.95, 0.95,	0.95, 0.95,	0.98, 0.96,	0.92, 0.94,	204, 159,
	0.95, 0.95	0.95, 0.95	0.98, 0.97	0.93, 0.93	157, 180



(a) 5 Epochs



(b) 10 Epochs



(c) 20 Epochs

FIGURE 9. Comparison between highway, LeNet, and Hi-Le models.

about the features influencing the model's predictions. As demonstrated in Figure 6, features learned by TCN along with haemoglobin and blood glucose level, emerge as the most significant contributors. The influence of each element is shown as a horizontal bar, with larger bars denoting more significance. Through a deeper knowledge of the ways in which features learned by TCN, haemoglobin, blood glucose influence the predictions of the HiTCL model, this visualization helps us to understand decision made by HiTCL model.

A comprehensive understanding of the influence of various variables on model predictions is offered by the SHAP beeswarm plot. Based on its SHAP value, each data point in this visualization represents a single prediction instance

and is arranged horizontally along the feature axis. The range of SHAP values connected to each feature and their related influence on the model's output are shown by the point distribution. Positively correlated features with the prediction are indicated by points concentrated near the top of the plot, whilst negatively correlated features are indicated by points clustered near the bottom. Furthermore, information on the variability and importance of each feature's contribution across several prediction instances can be gleaned from the spread and density of dots. From Figure 7, we can recognize important elements and comprehend their varied implications on model predictions.

The **dependence plot** of SHAP for haemoglobin and high glucose is shown in Figure 8 shows that changes

TABLE 13. Hyperparameters of highway, LeNet and Hi-Le model.

Common Parameters:	
Optimizer	Adam
Loss Function	Binary Cross_entropy
Metrics	Accuracy
Number of Epoch	5-20
Batch Size	64
Activation Function on Output Layer	Sigmoid
Highway Network:	
Hidden Layers	4
Activation Function in every Hidden layer	ReLU
Number of Neurons	32-64
LeNet-5	
kernel Size	3
Number of Neurons	6-64
Activation Function	tanh
Hi-Le:	
Hidden Layers in Highway Network	4
Number of Neurons	6-64
Activation Function	ReLU
Activation Function on Output Neuron	Sigmoid

TABLE 14. Highway, LeNet and Hi-Le models' architectures.

Model	Architecture
LeNet	input = Input(shape=(X_train.shape[1],)) Conv1D (neurons: 6, kernel_size: 3, activation: tanh) Max Pooling (pool_size=2) Conv1D (neurons:32, kernel_size: 3, activation: tanh) Max Pooling (pool_size=1) flatten() Dense (neurons:16, activation=tanh) Dense (64, activation=tanh) Dense (1, activation=sigmoid)
Highway Network	input = Input(shape=(X_train.shape[1],)) Transformation layer: Dense(neurons: 8, activation: sigmoid) Dense(neurons: 64, activation=ReLU) Dense(neurons: 64, activation=ReLU) Dense(neurons: 32, activation=ReLU) Dense(neurons: 32, activation=ReLU) Dense(neurons: 2, activation=Sigmoid)
Hi-Le Model	Conv1D (neurons: 6, kernel_size: 3, activation: tanh) Max Pooling (pool_size=2) Conv1D (neurons:32, kernel_size: 3, activation: tanh) Max Pooling (pool_size=1) flatten() Dense (neurons:16, activation=tanh) Dense (64, activation=tanh) Dense(neurons: 64, activation=ReLU) Dense(neurons: 64, activation=ReLU) Dense(neurons: 32, activation=ReLU) Dense(neurons: 32, activation=ReLU) Dense(neurons: 2, activation=Sigmoid)

to these specific features impact the output predictions of the model. The high blood glucose level feature's value is represented on the x-axis, and its impact on the model's prediction is indicated on the y-axis, which shows the SHAP value. The association between the feature's values and the SHAP values is depicted in the plot as a trend line. When the trend line for haemoglobin slopes upward, it shows that the model's prediction tends to grow with greater haemoglobin levels, whereas a downward slope reflects the opposite effect. In the same manner, a positive slope

TABLE 15. Hi-Le model results with SGD optimizer.

Models	Accuracy	F1-Score	Recall	Precision	E.T(s)
LeNet	0.59	0.36	0.85	0.22	83
Highway	0.72	0.76	0.72	0.71	29
Hi-Le	0.92	0.92	0.92	0.92	51

TABLE 16. Hi-Le model results with batch size 32.

Models	Accuracy	F1-Score	Recall	Precision	E.T(s)
LeNet	0.84	0.85	0.81	0.90	83
Highway	0.81	0.83	0.81	0.81	63
Hi-Le	0.94	0.96	0.92	0.94	103

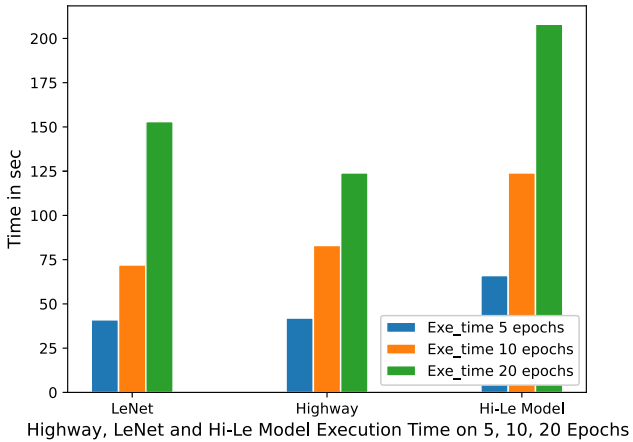


FIGURE 10. Execution time of highway, LeNet and Hi-Le model for 5, 10, 20 epochs.

for high glucose levels implies that higher levels influence predictions, whereas a negative slope denotes a negative relationship.

B. Hi-Le MODEL

Experimentation shows that the Hi-Le model gives more accurate and reliable results as compared to its individual models, Highway and LeNet. Evaluation metrics shows the superiority of Hi-Le model for both training and testing in Tables 9, 10, 11 and 12.

1) STRATIFIED K-FOLD CROSS VALIDATION

A variation on the standard k-FCV method called stratified k-FCV divides the data into k folds, while maintaining the proportion of each class in each fold. Because it guarantees that every fold accurately reflects the class distribution as a whole, this is especially helpful when working with balanced or imbalanced datasets. Every iteration uses a different fold as the testing set and the remaining folds as the training set, and the model is trained and evaluated k times. This procedure supports in evaluating the model's performance over various data subsets. We have split up the dataset into ten folds specifically for 10-FCV, and the training and evaluation procedures are carried out ten times over. The remaining nine folds are used for training, and a different fold is used

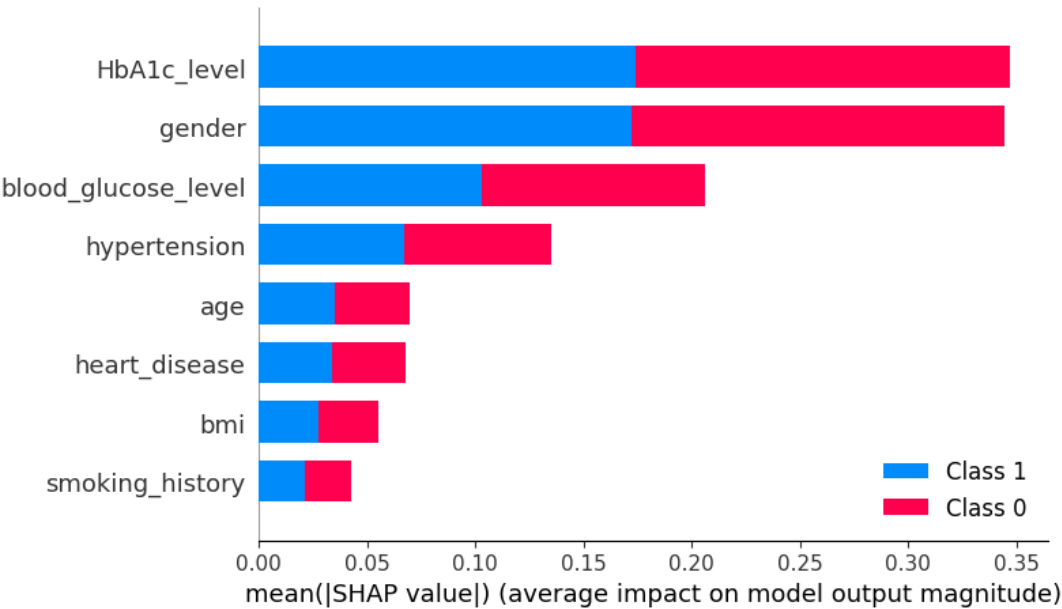


FIGURE 11. Summary plot for Hi_Le model.

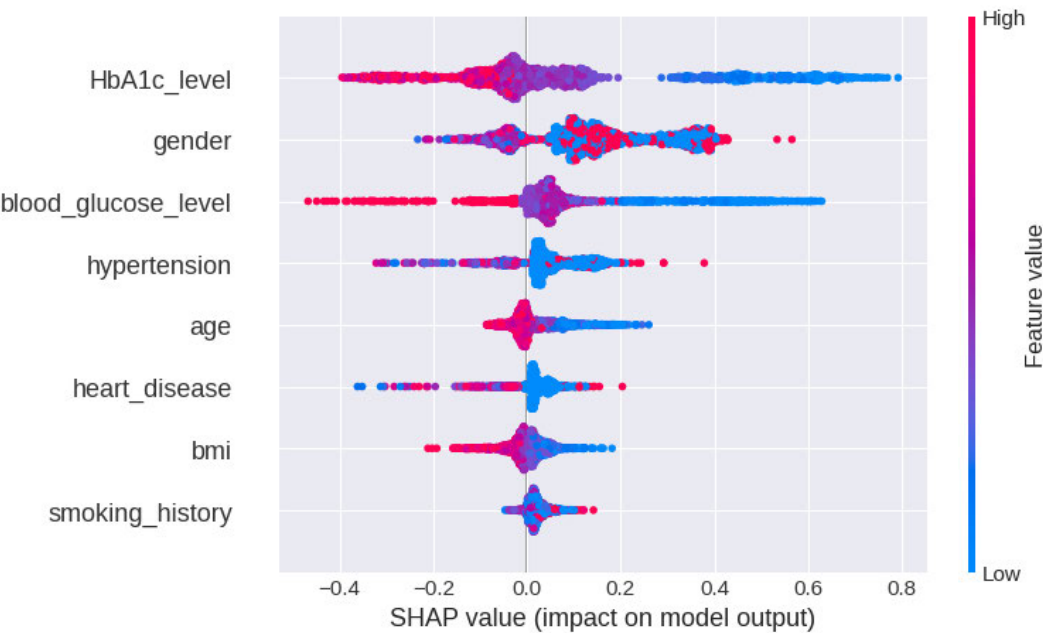


FIGURE 12. Beeswarm plot for Hi_Le model.

as the test set for each iteration. The average performance over all ten folds is then used to determine the model’s final performance.

LeNet and Highway work well together since their architectures are different and their combination works well, as shown in tables 10, 11, and 12. First bold value in table shows the execution of Hi-Le model on dataset and last bold value shows the average value of 10-FCV. The

highway model manages the information flow across layers, whereas the LeNet model is a convolutional model that excels at capturing the spatial associations of features in data. Thus, the Hi-Le model incorporates both spatial hierarchies and adaptive information processing as a result of their combination, and their graphical representation is shown in Figure 9. As there is a trade-off between model accuracy and execution time, we have achieved accurate prediction,

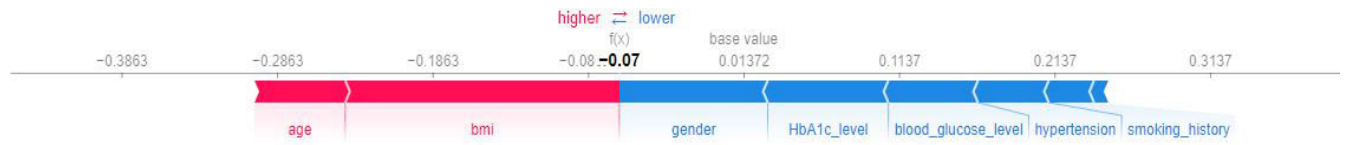


FIGURE 13. Force plot for Hi-Le model.

TABLE 17. List of abbreviations.

Abbreviation	Explanation
10-FCV	10-Fold Cross Validation
1D	1 Dimensional
DL	Deep Learning
DNN	Deep Neural Network
DPD	Diabetes Prediction Dataset
Hi-Le	Highway-LeNet
HiTCLe	Highway-TCN-LeNet
LSTM	Long Short Term Memory
MLP	Multi Layer Perceptron
ML	Machine Learning
NN	Neural Network
ProWSyn	Proximity-Weighted Synthetic Oversampling
ReLU	Rectified Linear Unit
SGD	Stochastic Gradient Descent
SHAP	Shapley Additive eXplanations
TCN	Temporal Convolutional Network

so the execution time of the Hi-Le model is higher than that of individuals, as shown in Figure 10. Tables 13 and 14 display the hyperparameters utilized in the individual models' entire architectures and the Hi-Le models' architectures.

2) ABLATION STUDY

A useful method for comprehending, evaluating, and enhancing DL models is ablation studies. Informed decisions regarding model architecture, feature selection, and algorithmic choices can be made by researchers with their assistance, as they offer a methodical and regulated approach to examining the influence of individual components. Comparisons of the Hi-Le model with its individual base models, highway and LeNet, are shown in tables 10, 11, and 12 on 5, 10, and 20 epochs. Results show a minor change in the results of models on different epochs. The experimentation was conducted by changing the optimizer from Adam to SGD, as shown in table 15. And Table 17 shows model results with batch size 32. From the results, it can be seen that our Hi-Le model's performance is good, but individual models are not working well with this combination of hyperparameters.

SHAP on Hi-Le Model: In this study, we have implemented SHAP on Hi-Le model as well. An essential part of the explainer is the masker. In order to compute SHAP values, it decides how to alter the input data. The kind of masker you use will rely on the type of data you have. The masker plays a crucial role in offering a reference point for evaluating the model's output when specific features are "masked" or assigned a fixed value.

In SHAP, a *summary plot* is a visual aid that offers an in-depth explanation of how each feature affects model predictions. It is an effective tool used for figuring out each feature contributes. The features are arranged in order of significance for affecting the model's output along the y-axis. Predictions are greatly affected by features located higher on the plot. Each color represents the feature value for that particular data point. The relationship between feature values and how they affect the model's prediction is made easier to understand. Figure 11 shows summary plot. The highest contributing feature in prediction is *HbA1c_level* where haemoglobin A1c is a measure of a person's average blood sugar level over the past 2-3 months. Figure 11 shows the contribution of both classes in diabetes prediction which is almost equal. So, we can say that both the classes play an equal role in diabetes prediction and haemoglobin A1c is influencing the model's prediction.

An extensive summary of feature importance and the distribution of SHAP values among features in a dataset can be obtained from a SHAP beeswarm plot. The width of each beeswarm, which denotes a feature, relates to the density of SHAP values and shows how different features affect model predictions. Features with wider sections in Figure 12 show higher variability in SHAP values, indicating a stronger impact on model predictions. Notably, the beeswarm plot's color variations can be used to depict the strength or direction of SHAP values; for example, blue indicates negative contributions and red indicates positive contributions. Red sections indicate cases where higher values positively contribute to predictions. haemoglobin A1c is highlighted as an example of a feature with significant impact.

A SHAP *force plot* is a detailed visualization that provides an in-depth understanding of how individual features contribute to the prediction of a specific instance. This plot is particularly useful for interpreting the decision-making process of a DL model for a given data point. Figure 13 shows the force plot for instance index number 265 represents features contribution to moving the model's output from the base value to the actual prediction for the instance. The amount of each feature's contribution is indicated by its length; longer arrows are indicative of features that have a bigger positive or negative effect on the prediction at that particular data point.

V. CONCLUSION

Diabetes is a chronic disease that works like a slow poison, and its early detection can reduce its effects on

individuals. First, we use the ProWSyn approach to balance the DPD dataset, which is well-known for having a high class imbalance. This helped to mitigate potential biases towards the majority class. We incorporate two ensemble approaches, blending and hybridization, into our framework. Our hybrid model, Hi-Le, achieves impressive results with an accuracy score of 94% and an F1-Score of 96% by combining the advantages of the highway and LeNet models. The results show that our proposed model outperforms the performance of the standalone DL models, proving the effectiveness of our hybrid strategy. Also, HiTCL, our blending model, combines the highway, TCN, and LeNet models and achieves a 94% accuracy and 94% F1-score. Both Hi-Le and HiTCL show potential in accurate and early diabetes prediction, which could lead to life-saving outcomes. It is also important to note that our methodology involves a thorough evaluation process that uses a 10-FCV strategy to guarantee the reliability of the performance measures of our models. Furthermore, our analysis' improved interpretability results from including SHAP illuminate the fundamental mechanisms guiding the model's predictions. Moreover, ablation research has been carried out to evaluate the specific contribution of every element in our ensemble models, offering important perspectives on their effectiveness and pointing out possible areas for development. By putting forward innovative ensemble methodologies and adopting innovative techniques for model evaluation and interpretation, this study contributes to the field of early diabetes detection.

APPENDIX

LIST OF ABBREVIATIONS

See Table 17.

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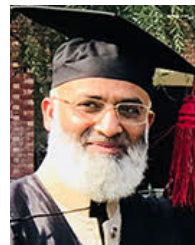
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